

MA4110/MA5020: Computational Methods for Fluid Flow

Tutorial Sheet 1

Classification of PDEs, Method of Characteristics, and Shock Formation

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1. For a general second-order PDE $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = G$, recall that the type is decided entirely by the discriminant $\Delta = B^2 - 4AC$. Classify each of the following as hyperbolic, parabolic, or elliptic in the regions of the (x, y) plane where they are defined, and where the type changes, identify the boundary curve separating the two regimes.

- (a) $x u_{xx} + u_{yy} = 0$
- (b) $u_{xx} + 2x u_{xy} + (1 - y^2) u_{yy} = 0$
- (c) $u_{xx} + 4u_{xy} + 4u_{yy} + 2u_x - u = 0$

2. Solve the initial value problem

$$\frac{\partial u}{\partial t} + (1 + x^2) \frac{\partial u}{\partial x} = 0, \quad x \in \mathbb{R}, \quad t > 0,$$

subject to $u(x, 0) = \frac{1}{1 + x^2}$.

- (a) Write down the characteristic equation and find the characteristic curve $x(t; x_0)$ passing through $(x_0, 0)$.
 - (b) Show that a characteristic starting at finite x_0 reaches $x \rightarrow +\infty$ in finite time, and find this transit time.
 - (c) Determine the explicit solution $u(x, t)$.
3. Consider $u_t + 2u_x = 0$ with initial data $u(x, 0) = u_0(x)$.
- (a) Find the characteristic line through the point $(x, t) = (5, 2)$, and state the domain of dependence of this point.
 - (b) If the initial data is perturbed only on the interval $x_0 \in [0, 1]$, describe (and sketch) the domain of influence of this interval, and shade the affected region of the (x, t) plane at time $t = 3$.
4. Consider the inviscid Burgers' equation $u_t + uu_x = 0$ with $u(x, 0) = u_0(x) = \frac{1}{1 + x^2}$.
- (a) Write the characteristic lines $x(t; x_0) = x_0 + u_0(x_0) t$.
 - (b) Compute $u'_0(x)$ and identify the regions of compression ($u'_0 < 0$) and expansion ($u'_0 > 0$).
 - (c) Find $\min_x u'_0(x)$ and hence the breaking time $t_s = -1/\min_x u'_0(x)$.
 - (d) Find the location x_s where the slope first becomes vertical.
5. For the inviscid Burgers' equation $u_t + uu_x = 0$ with smooth initial data $u(x, 0) = u_0(x)$, recall that characteristics are the straight lines $x(t; x_0) = x_0 + u_0(x_0) t$, and the solution first develops a vertical tangent (a shock) at the breaking time

$$t_s = \frac{-1}{\min_x u'_0(x)},$$

provided $\min_x u'_0(x) < 0$ somewhere; if $u'_0(x) \geq 0$ everywhere, no shock ever forms. For each initial profile below, find $u'_0(x)$, locate $\min_x u'_0(x)$ (and the point x_0^* where it occurs), and hence compute t_s . Where relevant, also give the breaking point $x_s = x_0^* + u_0(x_0^*) t_s$.

(a) $u_0(x) = -\sin(x)$

(b) $u_0(x) = -\tanh(x)$

(c) $u_0(x) = \begin{cases} 1, & x \leq 0 \\ 1-x, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases}$

(d) $u_0(x) = -x e^{-x^2}$

(e) $u_0(x) = \frac{2}{1+x^2}$

6. For each PDE and initial data below, find the characteristic curves $x(t; x_0)$, sketch a family of them (for several representative values of x_0) in the (x, t) half-plane $t \geq 0$, and use the sketch to answer the question posed.

(a) $u_t + u_x = 0$, $u(x, 0) = u_0(x)$. Sketch the characteristics for $x_0 = -2, -1, 0, 1, 2$. Since these are parallel straight lines, explain why the solution is just a rigid translation of u_0 , and mark on your sketch the point reached at $t = 3$ by the characteristic starting at $x_0 = -1$.

(b) $u_t + x u_x = 0$, $u(x, 0) = u_0(x)$. Show the characteristic ODE is $\frac{dx}{dt} = x$, solve it, and sketch the characteristics for $x_0 = -2, -1, -0.5, 0.5, 1, 2$. What do you notice about the characteristic through $x_0 = 0$, and about the spacing between neighboring characteristics as t increases?

(c) $u_t + t u_x = 0$, $u(x, 0) = u_0(x)$. Find $x(t; x_0)$ (note the characteristic speed now depends on t , not x or u), and sketch the resulting parabolic characteristic curves for $x_0 = -2, -1, 0, 1, 2$.

(d) $u_t + u u_x = 0$ with $u_0(x) = x$ for $x \in [0, 1]$ and $u_0(x) = 0$ otherwise. Sketch the characteristics for $x_0 = 0, 0.25, 0.5, 0.75, 1$, together with a few from the surrounding constant regions. Do any characteristics cross for $t > 0$? What does the fan of characteristics emanating from $x_0 = 0$ look like, and what kind of solution structure does it signal?

(e) $u_t + u u_x = 0$ with $u_0(x) = 1$ for $x < 0$ and $u_0(x) = 0$ for $x > 0$. Sketch the characteristics from both sides of $x_0 = 0$ and mark the point where they first intersect. Shade the region of the (x, t) plane that is left uncovered by any characteristic drawn from $t = 0$, and briefly note what must happen there once a shock is inserted.

7. Recall the hierarchy for first-order PDEs of the general form $F(x, t, u, u_x, u_t) = 0$:

- **Linear:** Linear in u, u_x, u_t , with coefficients depending only on (x, t) , i.e., $a(x, t)u_t + b(x, t)u_x + c(x, t)u = d(x, t)$.
- **Semilinear:** Linear in the highest derivatives (u_t, u_x) with coefficients depending only on (x, t) , while non-linearities enter only through u , i.e., $a(x, t)u_t + b(x, t)u_x = f(x, t, u)$.
- **Quasilinear:** Linear in the highest derivatives (u_t, u_x) , but their coefficients may depend on u as well as (x, t) , i.e., $a(x, t, u)u_t + b(x, t, u)u_x = f(x, t, u)$.
- **Fully Nonlinear:** Non-linear in the highest derivatives (u_t, u_x) , e.g., $(u_x)^2$, $\sqrt{1 + u_x^2}$, etc.

Classify each first-order PDE below into one of these four categories (**Linear**, **Semilinear**, **Quasilinear**, or **Fully Nonlinear**) and justify your choice:

(a) $u_t + c u_x = 0$, c a constant

- (b) $u_t + x u_x = t$
- (c) $u_t + u u_x = 0$ (Inviscid Burgers' equation)
- (d) $u_t + u_x = u^2$
- (e) $u_t + u^2 u_x = 0$
- (f) $(u_x)^2 + u_t = 0$ (Hamilton-Jacobi type equation)
- (g) $u_t + \sin(x) u_x + u = 0$
- (h) $u_t + \cos(u) u_x = 0$ (Traffic flow model with density-dependent velocity)
- (i) $a(x, t) u_x + b(x, t) u_t + c(x, t) u = f(x, t)$
- (j) $u_t + \sqrt{1 + (u_x)^2} = 0$ (Eikonal / geometric front propagation equation)